

Exercise 7

UNION & UNION ALL

Online_sales	Customer_name	Amount	Sales date
Sale_id	Alice	150	2025-01-12
1	Brian	250	2025-02-05
2	Carol	300	2025-03-10
3	Daniel	220	2025-04-15
4	Emma	180	2025-05-02
5			
Store sales			

Sales_id	Customer_name	Amount	Sales_date
1	Fiona	200	2025-01-20
2	Brian	250	2025-02-08
3	George	310	2025-03-25
4	Alice	150	2025-04-18
5	Henry	270	2025-05-05

① Unique Customer names

List all unique customer names from both tables using UNION

O.C : customer name

```
SELECT customer_name FROM Online sales
UNION
```

```
SELECT customer_name FROM Store sales;
```

Alice
 Brian
 Carol
 Emma
 Fiona
 George
 Henry

② All customers (including all duplicates)

SELECT customer_name from online sales

UNION ALL

SELECT Customer_name from store sales;

Customer name

Alice

Brian

Carol

Emma

Fiona

Brian

George

Alice

Henry

③ Show all unique sales dates from both tables in ascending

SELECT sale_date FROM online sales

UNION

SELECT sale_date from store sales

ORDER BY sale_date ASC;

Sale_date

2025-01-12

2025-01-20

2025-02-05

2025-02-08

2025-03-10

2025-03-25

2025-04-15

2025-04-18

2025-05-02

2025-05-05

④ All sale dates (including duplicates)

SELECT sale date from Online sales

UNION ALL

SELECT sale date from store sales

Sale date

2025-01-12

2025-02-05

2025-03-10

2025-04-15

2025-05-02

2025-01-20

2025-02-08

2025-03-25

2025-04-18

2025-05-05

⑤ High-value customers

Combine both tables & list unique customers who made purchases greater than 250

SELECT customer_name, amount from online_sales WHERE
amount > 250

UNION

SELECT customer_name, amount from store_sales WHERE amount
> 250;

Customer name	Amount
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Carol

300

George

310

Henry

270

⑥ Combined sales data

Combine all records from both tables using UNION ALL

```
SELECT customer_name, amount, sale_date FROM online_sales  
UNION ALL
```

```
SELECT customer_name, amount, sale_date FROM store_sales
```

Customer name	Amount	Sale date
Alice	150	2025-01-12
Brian	250	2025-02-05
Carol	300	2025-03-10
Daniel	220	2025-04-15
Emma	180	2025-05-02
Fiona	200	2025-06-20
Brian	250	2025-02-08
George	310	2025-03-25
Alice	150	2025-04-18
Henry	270	2025-05-05

⑦ Add sale source label — indicate was made online or Store

```
SELECT customer_name, amount, sale_date 'Online' AS  
source FROM online_sales
```

```
UNION ALL
```

```
SELECT customer_name, amount, sale_date 'Store' AS  
source FROM store_sales;
```


Customer-name	amount	sale-date	Source
Alice	150	2025-01-12	Online
Brian	250	2025-02-03	Online
Carol	300	2025-03-0	Online
Daniel	220	2025-04-15	Online
Emma	180	2025-05-02	Online
Fiona	200	2025-01-20	Store
Brian	250	2025-02-08	Store
George	310	2025-03-25	Store
Alice	150	2025-04-18	Store
Henry	210	2025-05-05	Store

2 Customers Appearing in Both tables
(Hint : UNION ALL, GROUP BY & HAVING)

* Occurance

SELECT customer-name, COUNT(customer-name) AS occurrences
FROM

SELECT customer-name FROM Online-sales
UNION ALL

SELECT customer-name FROM store-sales
GROUP BY customer-name
HAVING COUNT(customer-name) > 1;

Customer-name	Occurrences
Alice	2
Brian	2

⑨ Total combined Sales

Combine both tables using UNION ALL and calculate the total sales amount across both

```
SELECT sum(Amount) AS Total amount  
SELECT amount from online sales  
UNION ALL  
SELECT amount from Store sales  
FROM all sales;
```

Total amount
2300